

Lab 2 – Linear Data Structures

Release: 18 Sep 2026 (Friday, Week 5)

Due: Submission through GitHub by Week 6 Sunday, 27 Sep 2026, 11pm

Instructions:

1. This is an individual lab. You will be awarded class participation points for the labs' component based on the outcome of the test cases results on GitHub, not on the efficiency of the solution (you are encouraged to explore the most efficient approach for your own learning purposes)
2. You can discuss the general approach to solve the question but you should not be sharing your code with anyone.
3. Unless specifically stated in the question, you are allowed to create additional methods in the classes.
4. Please ensure that all your codes can be compiled.
5. The template repository for Lab 2 is at <https://github.com/cookiemo729/cs201lab2-student-template>
6. Remember to enable GitHub workflow at the Actions tab for your pushes to be autograded. Refer to the guide "CS201_AutoGrader_Student_Guide.pdf" (uploaded on eLearn with Lab 1) if you need to recall how to access the repository and submit on GitHub.
7. **You must submit your solution through GitHub. You may submit your work multiple times to GitHub. Each submission automatically triggers the Autograder, allowing you to receive feedback and improve your solution.**
8. Lab2Test.java (in the GitHub repository) contains the tester code that's used by the autograder. You could use that to locally test your code before pushing your solution to GitHub.
9. **No participation points** will be awarded for **submissions made after the due date / time**.

Q1 – Swapping Elements in a Single Linked List

An implementation of a Single Linked List has been given in SinglyLinkedList.java. Study the code and complete the following requirements in SinglyLinkedList.java to extend the functionality of the linked list. Please ensure the head and tail nodes are set accordingly if they are impacted by the method.

- Write a method named **swap** that swap the sequence of all the elements whereby the biggest value will be swapped with the lowest value, the second biggest value will be swapped with the second lowest value....
- You can **assume all the values are unique**.
- You **should not be** making changes (e.g. adding new getter methods) to the Node class within SinglyLinkedList or to update the values of the element object within Node class directly.
- You could try to include more test cases (e.g. using a sequence of (1 to 1000) or (1 to 10000) to see how your code will respond to a bigger input size

Before Swap :

Linked List : 3 2 1

First Element : 3

Last Element : 1

After Swap :

Linked List : 1 2 3

First Element : 1

Last Element : 3

Before Swap :

Linked List : 5 1 2 3

First Element : 5

Last Element : 3

After Swap :

Linked List : 1 5 3 2

First Element : 1

Last Element : 2

Before Swap :

Linked List : 5 3 4 1 2

First Element : 5

Last Element : 2

After Swap :

Linked List : 1 3 2 5 4

First Element : 1

Last Element : 4